

A broken stick and a Feynman diagram

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Abstract

Break a stick at five random points and use the six pieces, in a fixed order, as the edges of a tetrahedron. How often does a tetrahedron exist? We show that the answer is a Feynman integral: up to an explicit constant it is the three-loop tetrahedron vacuum diagram in four dimensions, with equal masses and every propagator raised to the power $3/2$. The same argument works in every dimension, with the complete graph K_{n+1} in place of the tetrahedron, and for $n = 2$ it gives the classical answer $1/4$ for triangles. Two independent computations of the tetrahedron integral agree to within 10^{-12} , giving $0.0125749944167\dots$, which corrects a value of $1/79$ reported from simulation. The algebraic steps are checked in the Lean proof assistant.

1 Six pieces of a stick

Take a stick of length 1 and break it at five points chosen independently and uniformly at random. Label the six pieces in order along the stick and lay them on the six edges of a tetrahedron in a fixed way, as in Figure 1. Write p_3 for the probability that some tetrahedron in space has exactly these edge lengths.

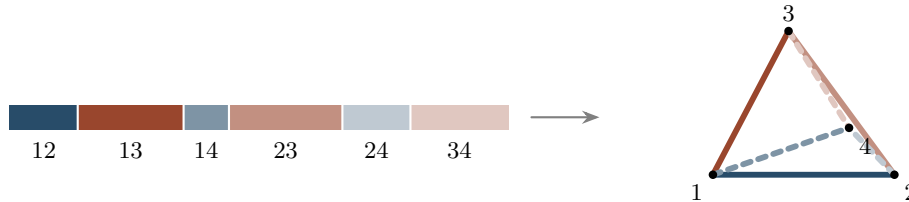


Figure 1: A stick broken at five random points. The pieces go to fixed edges of a tetrahedron, here 12, 13, 14, 23, 24, 34 in order. The question is whether a tetrahedron with these edge lengths exists.

The one-dimension-lower version is a classic. Stop and try it before reading on: break a stick at two random points; how likely are the three pieces to form a triangle? The answer is $1/4$, because the triangle exists exactly when no piece is longer than $1/2$, and each of the three ways to fail cuts off a quarter of the triangle of possible lengths.

For the tetrahedron the four faces must satisfy the triangle inequality, but that is not enough. Douglas Zare observed on MathOverflow that the twelve triangle inequalities cut out a polytope with $1/54$ of the possible lengths [2]. Inside it a sixth-degree polynomial in the lengths, the Cayley–Menger determinant, must also have the right sign, and this extra condition removes about a third of the polytope. Benjamin Dickman asked the question in 2013 [1, 3], and surveys of broken-stick problems pose it as the natural next case [8, 13]. No exact value has been found, and the only numbers so far are numerical estimates, one of which, $1/79$, is reported in the comments of [1].

This note does not find a closed form either. It shows instead why the number is hard, by identifying it with an object physicists have studied for decades. Write $N = n(n + 1)/2$ for

the number of edges of an n -simplex and U for the first Symanzik polynomial of the complete graph K_{n+1} (defined in Section 3): for each spanning tree, multiply the variables of the edges *not* in the tree, and add over all spanning trees. Let p_n be the probability that N pieces of a randomly broken stick, laid on the edges of an n -simplex in a fixed order, are its edge lengths.

Theorem 1. *For every $n \geq 2$,*

$$p_n = C_n \int_{\mathbb{R}_+^N} \frac{\prod_e \beta_e^{(n-2)/2} e^{-\sum_e \beta_e}}{U(\beta)^{(n+1)/2}} d\beta, \quad C_n = \frac{2^{N-n}}{\pi^{N/2}} \Gamma_n\left(\frac{n+1}{2}\right), \quad (1)$$

where $\Gamma_n(a) = \pi^{n(n-1)/4} \prod_{j=0}^{n-1} \Gamma(a - j/2)$ is the multivariate gamma function.

For the tetrahedron, $C_3 = 4/\pi$ and the integrand is $\prod_e \beta_e^{1/2} e^{-\sum \beta_e} / U(\beta)^2$. In the language of quantum field theory, Section 4 shows that

$$p_3 = \frac{\pi^2}{16} T, \quad T = \int \frac{d^4 k_1 d^4 k_2 d^4 k_3}{\pi^6} \prod_{6 \text{ lines}} \frac{1}{(q^2 + 1)^{3/2}},$$

the tetrahedron vacuum diagram of Figure 2, with three loops in four Euclidean dimensions. Section 5 computes $p_3 = 0.0125749944\dots$ by two independent methods and p_4 by Monte Carlo, and Section 6 says what is still open, including the original question, in which the pieces may be laid on the edges in any order.

2 From a broken stick to a Gram matrix

Three classical facts turn the probability into an integral.

Exponential pieces. If $E_1, \dots, E_N$ are independent with the exponential distribution of mean 1, then $(E_1, \dots, E_N) / \sum E_i$ has the distribution of the pieces of a stick broken at $N-1$ independent uniform points [7, Chapter V, Theorem 2.2]. Whether given lengths form a simplex does not change when they are all multiplied by the same positive number. So p_n is also the probability that N independent exponential lengths ℓ_e form a simplex:

$$p_n = \int_{\text{simplex lengths}} \prod_e e^{-\ell_e} d\ell.$$

Schoenberg's criterion. Number the vertices $0, 1, \dots, n$ and write ℓ_{ij} for the length on edge ij . Put

$$G_{ij} = \frac{1}{2}(\ell_{0i}^2 + \ell_{0j}^2 - \ell_{ij}^2) \quad (1 \leq i, j \leq n, \ell_{ii} = 0). \quad (2)$$

If the simplex exists with vertex 0 at the origin and the other vertices at $x_1, \dots, x_n$, then $G_{ij} = x_i \cdot x_j$ is their Gram matrix. Schoenberg proved the converse: the lengths belong to points of $\mathbb{R}^n$ exactly when G is positive semidefinite [12] (see also [10, Section 2]), and the points span a genuine n -simplex exactly when G is positive definite, since $\det G = (n!V)^2$ with V the volume. For the triangle, $G = \begin{pmatrix} \ell_{01}^2 & * \\ * & \ell_{02}^2 \end{pmatrix}$ with $*$ = $\frac{1}{2}(\ell_{01}^2 + \ell_{02}^2 - \ell_{12}^2)$, and $\det G > 0$ is Heron's formula in disguise.

Changing variables. Formula (2) is a linear bijection between the squared lengths $q_e = \ell_e^2$ and the symmetric matrices G . The diagonal of G copies $q_{01}, \dots, q_{0n}$, and each off-diagonal entry contributes a factor $-\frac{1}{2}$, so $dG = 2^{-n(n-1)/2} dq$, where dG is Lebesgue measure on the entries G_{ij} , $i \leq j$. Since $dq_e = 2\ell_e d\ell_e$ and $N - n(n-1)/2 = n$,

$$p_n = 2^{-n} \int_{G \succ 0} \prod_e \frac{e^{-\ell_e}}{\ell_e} dG, \quad (3)$$

where each ℓ_e is now the function of G given by (2). Read (3) as: integrate over all simplices with vertex 0 at the origin, described by their Gram matrices, with weight $\prod_e e^{-\ell_e}/\ell_e$. The cone $G \succ 0$ is simple. The weight is the hard part.

3 From a Gram matrix to a graph polynomial

The weight $e^{-\ell}/\ell$ is a function of ℓ , while G is linear in ℓ^2 . The following identity bridges the two. It is a Gaussian integral in disguise, and is often called the Schwinger representation.

Lemma 1. For $\ell > 0$, $\frac{e^{-\ell}}{\ell} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} \exp\left(-\ell^2 t - \frac{1}{4t}\right) dt$.

Proof. Put $t = s^2$. The integral becomes $2 \int_0^\infty \exp(-\ell^2 s^2 - 1/(4s^2)) ds$. Complete the square: $\ell^2 s^2 + 1/(4s^2) = (\ell s - 1/(2s))^2 + \ell$, so the integral is $2e^{-\ell} \int_0^\infty e^{-(\ell s - 1/(2s))^2} ds$, and it remains to show that the last integral is $\sqrt{\pi}/(2\ell)$. The substitution $s \mapsto 1/(2\ell s)$ maps $(0, \infty)$ to itself and fixes $\ell s - 1/(2s)$ up to sign, so averaging the integral with its image gives $\int_0^\infty e^{-(\ell s - 1/(2s))^2} ds = \frac{1}{2} \int_0^\infty e^{-(\ell s - 1/(2s))^2} \left(1 + \frac{1}{2\ell s^2}\right) ds = \frac{1}{2\ell} \int_{\mathbb{R}} e^{-u^2} du = \frac{\sqrt{\pi}}{2\ell}$, with $u = \ell s - 1/(2s)$. This is the Cauchy–Schlömilch transformation [4]. $\square$

Apply the lemma to each of the N factors in (3), with a variable t_e for each edge. The exponent collects into $-\sum_e t_e \ell_e^2$, and this is linear in G :

$$\sum_e t_e \ell_e^2 = \text{tr}(L_t G), \quad (4)$$

where L_t is the weighted Laplacian of K_{n+1} with weight t_{ij} on edge ij , with the row and column of vertex 0 removed. (For points x_i with $x_0 = 0$, both sides equal $\sum_{i < j} t_{ij} |x_i - x_j|^2$; as (4) is a polynomial identity in G , it holds for every symmetric G .) All the integrands are positive, so the order of integration may be changed, and the inner integral over G is the matrix version of $\int_0^\infty e^{-\lambda g} dg = 1/\lambda$:

$$\int_{G \succ 0} e^{-\text{tr}(LG)} dG = \Gamma_n\left(\frac{n+1}{2}\right) (\det L)^{-(n+1)/2} \quad (L \succ 0), \quad (5)$$

Siegel's matrix gamma integral; for $L = I$ it is the definition of Γ_n , and the general case follows by the substitution $G \mapsto L^{-1/2} G L^{-1/2}$, whose Jacobian is $(\det L)^{-(n+1)/2}$ [11, Section 2.1]. Its value depends on t only through $\det L_t$, and Kirchhoff's matrix-tree theorem [9] says what that is: the sum, over all spanning trees T of K_{n+1} , of $\prod_{e \in T} t_e$. Call this $\Psi(t)$. Putting the pieces together,

$$p_n = 2^{-n} \pi^{-N/2} \Gamma_n\left(\frac{n+1}{2}\right) \int_{\mathbb{R}_+^N} \prod_e t_e^{-1/2} e^{-1/(4t_e)} \Psi(t)^{-(n+1)/2} dt.$$

Two small changes give Theorem 1. Substitute $t_e = 1/(4\beta_e)$, so that $t_e^{-1/2} e^{-1/(4t_e)} dt_e = \frac{1}{2} \beta_e^{-3/2} e^{-\beta_e} d\beta_e$. And replace Ψ by its dual: a spanning tree of K_{n+1} has n edges, so its complement has $L = N - n$, and

$$\Psi(t) = \left(\prod_e t_e\right) U(1/t), \quad U(\beta) = \sum_T \prod_{e \notin T} \beta_e. \quad (6)$$

With $t = 1/(4\beta)$ this reads $\Psi = 4^{-n} U(\beta) / \prod_e \beta_e$, because U is homogeneous of degree $N - n$. Collecting powers, $\Psi^{-(n+1)/2}$ contributes $4^N \prod_e \beta_e^{(n+1)/2} U(\beta)^{-(n+1)/2}$ (as $n(n+1)/2 = N$), the N substitutions contribute $2^{-N} \prod_e \beta_e^{-3/2}$, and the product is $2^N \prod_e \beta_e^{(n-2)/2} U(\beta)^{-(n+1)/2}$. This proves (1), with $C_n = 2^{N-n} \pi^{-N/2} \Gamma_n((n+1)/2)$.

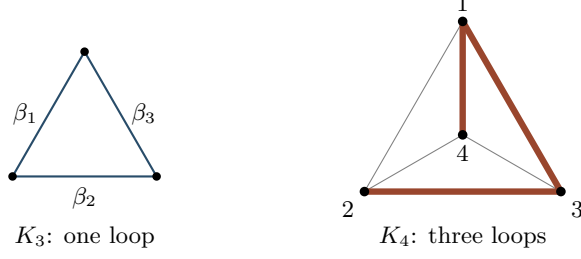


Figure 2: Left: the triangle K_3 ; its three spanning trees each leave out one edge, so $U = \beta_1 + \beta_2 + \beta_3$. Right: the tetrahedron K_4 drawn flat, as a vacuum Feynman diagram with four vertices, six lines and three loops. One of its 16 spanning trees is drawn heavy; the three edges left out give the monomial $\beta_{12}\beta_{34}\beta_{24}$ of U .

The triangle again. For $n = 2$ there are $N = 3$ edges, every spanning tree leaves out exactly one edge, and $U(\beta) = \beta_1 + \beta_2 + \beta_3$. The power $(n - 2)/2$ is 0, so (1) reads

$$p_2 = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}_+^3} \frac{e^{-(\beta_1 + \beta_2 + \beta_3)}}{(\beta_1 + \beta_2 + \beta_3)^{3/2}} d\beta = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{s^2}{2} s^{-3/2} e^{-s} ds = \frac{\Gamma(3/2)}{2\sqrt{\pi}} = \frac{1}{4},$$

grouping the β by their sum s (the set $\beta_1 + \beta_2 + \beta_3 = s$ has area element $s^2/2$). The long road through Gram matrices and spanning trees ends where the short argument of Section 1 did.

The simplex form. The integrand of (1) without the exponential is homogeneous of degree $N(n - 2)/2 - (N - n)(n + 1)/2 = -N/2$. Grouping β by its sum in the same way gives

$$p_n = C_n \Gamma(N/2) \int_{\Delta} \frac{\prod_e \sigma_e^{(n-2)/2}}{U(\sigma)^{(n+1)/2}} d\sigma, \quad (7)$$

over the simplex $\Delta = \{\sigma_e \geq 0, \sum \sigma_e = 1\}$ with its Lebesgue measure of total mass $1/(N - 1)!$. This is the form used for computing.

4 The tetrahedron as a Feynman diagram

In a Euclidean quantum field theory with one particle of mass 1, a vacuum diagram is a graph whose lines carry momenta $q_e \in \mathbb{R}^D$, conserved at each vertex, so that the free momenta are one $k \in \mathbb{R}^D$ per independent loop. A line with propagator power ν contributes $(q_e^2 + 1)^{-\nu}$. Writing each factor as $\Gamma(\nu)^{-1} \int_0^\infty \alpha^{\nu-1} e^{-\alpha(q_e^2 + 1)} d\alpha$ and doing the Gaussian integral over the loop momenta gives the standard parametric representation [5, eqs. (7)–(9) and Section 4]:

$$\int \prod_{\text{loops}} \frac{d^D k}{\pi^{D/2}} \prod_e \frac{1}{(q_e^2 + 1)^\nu} = \frac{1}{\Gamma(\nu)^N} \int_{\mathbb{R}_+^N} \frac{\prod_e \alpha_e^{\nu-1} e^{-\sum \alpha_e}}{U(\alpha)^{D/2}} d\alpha,$$

with the same U , the sum over spanning trees of the products of the omitted variables. Comparing with Theorem 1: $\nu - 1 = (n - 2)/2$ and $D/2 = (n + 1)/2$.

Corollary 1. $p_n = C_n \Gamma(n/2)^N T_n$, where T_n is the vacuum diagram of the complete graph K_{n+1} in dimension $D = n + 1$, with all masses 1 and every propagator raised to the power $n/2$.

For the tetrahedron, $C_3 \Gamma(3/2)^6 = (4/\pi)(\pi^3/64) = \pi^2/16$, which is the formula $p_3 = \pi^2 T/16$ of the introduction. Table 1 lists the first three cases. The tetrahedron diagram with equal masses

n	lines N	loops $N - n$	dimension D	power ν	p_n
2	3	1	3	1	1/4
3	6	3	4	3/2	0.0125749944...
4	10	6	5	2	1.0584×10^{-4} (Monte Carlo)

Table 1: The diagram behind each p_n . For the triangle, $T_2 = \int d^3k/\pi^{3/2} (k^2 + 1)^{-3} = \Gamma(3/2)/\Gamma(3) = \sqrt{\pi}/4$ and $C_2 = 1/\sqrt{\pi}$, so $p_2 = 1/4$ once more.

has a long history in particle physics, where it appears at three loops. With every power equal to 1 in four dimensions, and masses 0 or 1, Broadhurst reduced its finite part to a few constants built from the sixth root of unity, such as $\zeta(3)$ and the Clausen value $\text{Cl}_2(\pi/3)$ [6]. The power 3/2 that the broken stick asks for is not among the cases treated there, and we have found no evaluation of it.

5 Numbers

The tetrahedron. Two computations that share only the reduction to exponential lengths give

$$p_3 = 0.0125749944167\dots,$$

agreeing to within 10^{-12} .

The first works in the lengths. Hold the two faces that meet along edge 12 and rotate one about that edge, as in Figure 3. The opposite edge 34 then takes every length between its values in the two flat positions, and no others, so the sixth exponential length can be integrated out exactly. Scaling out the length of edge 12 leaves a four-dimensional integral over the positions of the two apexes. In elliptic coordinates its integrand is smooth except where the two apexes project to the same point, and polar coordinates centred there with Gauss quadrature give 0.0125749944167; raising the inner order from 24 to 40 and the outer from 16 to 24 moves the result by less than 10^{-13} .

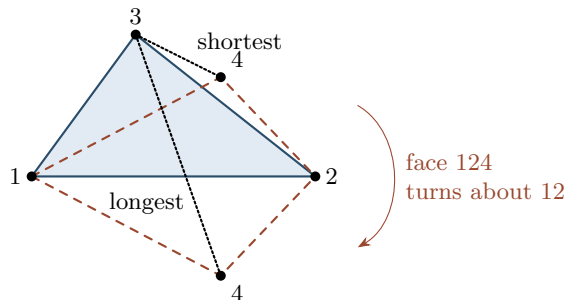


Figure 3: Two faces hinged on edge 12. As the face 124 turns, the length $|x_3x_4|$ runs over an interval whose ends are the two flat positions (dashed). A tetrahedron exists exactly when the sixth length lies inside this interval.

The second integrates the simplex form (7) with $n = 3$. Splitting Δ into six sectors according to its largest coordinate and putting $\sigma = s^2$ in the others removes every singularity, and product Gauss rules of orders 24, 28, ..., 44 give 0.0125749944383, ... 44216, ... 44180, ... 44171, ... 44168, ... 44168, settling on the same value from above.

A plain Monte Carlo simulation of the original question, testing the Cayley–Menger condition for 3×10^8 random sticks, gives 0.012576 ± 0.000006 . The value $1/79 = 0.0126582$ reported in [1]

is 0.66% too high, which is 13 standard errors of that simulation. Against Zare’s polytope, of volume $1/54$, the Cayley–Menger condition keeps $54p_3 \approx 68\%$.

The 4-simplex. Monte Carlo on both sides of Theorem 1 for $n = 2, 3, 4$, with 2×10^7 samples each, agrees within one standard error in every case. For $n = 4$ the integral side gives $p_4 = 1.0584 \times 10^{-4}$ with standard error 2×10^{-8} , and the direct simulation of ten exponential lengths gives $(1.069 \pm 0.023) \times 10^{-4}$.

Any order. The question as Dickman asked it [1] lets the six pieces be placed on the edges in whichever order works. Up to the symmetries of the tetrahedron there are 30 essentially different orders, and the event is the union of their 30 events. Randomised quasi-Monte Carlo over this union gives 0.0652818 ± 0.0000004 , in agreement with the estimate 0.0652823 ± 0.0000113 by Sasha on Mathematics Stack Exchange [3].

6 What is left

No closed form is known for p_3 or for the unordered probability. Searches for integer relations with three or four terms and coefficients up to 200, between p_3 (to eleven digits) and constants such as π , π^2 , $\zeta(3)$, $\log 2$, $\log 3$, $\sqrt{2}$, $\sqrt{3}$, Catalan’s constant and $\text{Cl}_2(\pi/3)$, found none. That is weak evidence at this precision. Theorem 1 at least places the problem: a closed form for p_3 is a closed form for an equal-mass three-loop diagram with half-integer powers, and the methods that evaluate such diagrams [6, 5] are the natural tools. The unordered probability is a union of 30 such events and is unlikely to be simpler.

Verification. The algebra behind Theorem 1 for the triangle and the tetrahedron is checked in Lean 4 with Mathlib [14], using only the standard axioms: the map (2) and its determinant, identity (4), the count of 16 spanning trees of K_4 , the matrix-tree expansion of $\det L_t$ for K_3 and K_4 , the duality (6), the substitution $t_e = 1/(4\beta_e)$, the homogeneity degree, the constants $C_2 = 1/\sqrt{\pi}$, $C_3 = 4/\pi$ and $C_3\Gamma(3/2)^6 = \pi^2/16$, and $p_2 = 1/4$. The analytic inputs (the exponential representation of a broken stick, Schoenberg’s criterion, Lemma 1 and Siegel’s integral) are quoted. Python programs compute p_3 by the two methods of Section 5, test Theorem 1 by Monte Carlo for $n = 2, 3, 4$, and assert every number stated here; deliberately wrong variants of the programs fail those checks. A C program computes the unordered probability. An independent program, sharing no code with the others, rechecks the values, Zare’s $1/54$, and the unordered probability over all 720 orders. Everything is at <https://github.com/jvvk/mathematics/tree/main/broken-stick-feynman>.

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